

Under the **Fréchet distribution** with parameters (A, α) , we recall:

$$f(x; \beta) = A\alpha(Ax)^{-(1+\alpha)} \exp(-(Ax)^{-\alpha}), \quad x > 0,$$

$$F(x; \beta) = \exp(-(Ax)^{-\alpha}).$$

✓ Likelihood in the LDM model under Fréchet

In the LDM setting, the arguments of f and F become:

$$x_{l_n} - \theta t.$$

Substituting f and F into your likelihood:

Full likelihood

$$\begin{aligned} \mathcal{L}(\theta; A, \alpha) &= \prod_{n=1}^{N_T} f(x_{l_n} - \theta l_n) \\ &\times \prod_{n=1}^{N_T-1} \prod_{t=l_n+1}^{l_{n+1}-1} F(x_{l_n} - \theta t) \\ &\times \left(\prod_{t=l_{N_T}+1}^T F(x_{l_{N_T}} - \theta t) \right)^{1_{l_{N_T} < T}}. \end{aligned}$$

✓ Substitute Fréchet f and F

Density terms:

$$f(x_{l_n} - \theta l_n) = A\alpha [A(x_{l_n} - \theta l_n)]^{-(1+\alpha)} \exp(-A(x_{l_n} - \theta l_n)^{-\alpha})$$

CDF terms:

$$F(x_{l_n} - \theta t) = \exp(-A(x_{l_n} - \theta t)^{-\alpha})$$

✓ Final expression

$$\begin{aligned}\mathcal{L}(\theta; A, \alpha) &= (A\alpha)^{N_T} \prod_{n=1}^{N_T} [A(x_{l_n} - \theta l_n)]^{-(1+\alpha)} \\ &\times \exp \left(- \sum_{n=1}^{N_T} [A(x_{l_n} - \theta l_n)]^{-\alpha} \right) \\ &\times \exp \left(- \sum_{n=1}^{N_T-1} \sum_{t=l_n+1}^{l_{n+1}-1} [A(x_{l_n} - \theta t)]^{-\alpha} \right) \\ &\times \exp \left(- 1_{l_{N_T} < T} \sum_{t=l_{N_T}+1}^T [A(x_{l_{N_T}} - \theta t)]^{-\alpha} \right).\end{aligned}$$

✓ Equivalent compact exponential form

$$\begin{aligned}\mathcal{L}(\theta; A, \alpha) &= (A\alpha)^{N_T} \prod_{n=1}^{N_T} [A(x_{l_n} - \theta l_n)]^{-(1+\alpha)} \\ &\times \exp \left(- \sum_{n=1}^{N_T} \sum_{t=l_n}^{l_{n+1}-1} [A(x_{l_n} - \theta t)]^{-\alpha} \right)\end{aligned}$$

(with $l_{N_T+1} = T + 1$ convention if you want full compactness).

✓ Log-likelihood (useful for inference)

$$\begin{aligned}\ell(\theta; A, \alpha) &= N_T \log(A\alpha) - (1 + \alpha) \sum_{n=1}^{N_T} \log(A(x_{l_n} - \theta l_n)) \\ &\quad - \sum_{n=1}^{N_T} (A(x_{l_n} - \theta l_n))^{-\alpha} \\ &\quad - \sum_{n=1}^{N_T-1} \sum_{t=l_n+1}^{l_{n+1}-1} (A(x_{l_n} - \theta t))^{-\alpha} \\ &\quad - 1_{l_{N_T} < T} \sum_{t=l_{N_T}+1}^T (A(x_{l_{N_T}} - \theta t))^{-\alpha}.\end{aligned}$$

↓

✓ Notation (to simplify formulas)

Define for any n, t :

$$y_{n,t} = x_{l_n} - \theta t, \quad z_{n,t} = A y_{n,t}$$

The log-likelihood is:

$$\ell(\theta, A, \alpha) = N_T \log(A\alpha) - (1 + \alpha) \sum_{n=1}^{N_T} \log(A y_{n,l_n}) - \sum_{n=1}^{N_T} \sum_{t \in \mathcal{I}_n} (A y_{n,t})^{-\alpha}$$

where:

$$\mathcal{I}_n = \begin{cases} \{l_n, \dots, l_{n+1} - 1\}, & n < N_T \\ \{l_{N_T}, \dots, T\}, & n = N_T \end{cases}$$

✓ 1. First-order derivatives

♦ w.r.t. θ

$$\frac{\partial \ell}{\partial \theta} = (1 + \alpha) \sum_{n=1}^{N_T} \frac{l_n}{y_{n,l_n}} - \alpha A \sum_{n=1}^{N_T} \sum_{t \in \mathcal{I}_n} t (z_{n,t})^{-\alpha-1}$$

♦ w.r.t. A

$$\frac{\partial \ell}{\partial A} = -\frac{\alpha N_T}{A} + \alpha \sum_{n=1}^{N_T} \sum_{t \in \mathcal{I}_n} y_{n,t} (z_{n,t})^{-\alpha-1}$$

♦ w.r.t. α

$$\frac{\partial \ell}{\partial \alpha} = \frac{N_T}{\alpha} - \sum_{n=1}^{N_T} \log(A y_{n,l_n}) + \sum_{n=1}^{N_T} \sum_{t \in \mathcal{I}_n} (z_{n,t})^{-\alpha} \log(z_{n,t})$$

✓ 2. Second-order derivatives

◆ $\partial^2/\partial\theta^2$

$$\frac{\partial^2 \ell}{\partial \theta^2} = +(1 + \alpha) \sum_{n=1}^{N_T} \frac{l_n^2}{y_{n,l_n}^2} - \alpha(\alpha + 1) A^2 \sum_{n=1}^{N_T} \sum_{t \in \mathcal{I}_n} t^2 (z_{n,t})^{-\alpha-2}$$

◆ $\partial^2/\partial A^2$

$$\frac{\partial^2 \ell}{\partial A^2} = \frac{\alpha N_T}{A^2} - \alpha(\alpha + 1) \sum_{n=1}^{N_T} \sum_{t \in \mathcal{I}_n} y_{n,t}^2 (z_{n,t})^{-\alpha-2}$$

◆ $\partial^2/\partial\alpha^2$

$$\frac{\partial^2 \ell}{\partial \alpha^2} = -\frac{N_T}{\alpha^2} - \sum_{n=1}^{N_T} \sum_{t \in \mathcal{I}_n} (z_{n,t})^{-\alpha} [\log(z_{n,t})]^2$$

◆ Cross derivatives

θ - A

$$\frac{\partial^2 \ell}{\partial \theta \partial A} = -\alpha \sum_{n,t} t(z_{n,t})^{-\alpha-1} + \alpha(\alpha+1)A \sum_{n,t} t y_{n,t} (z_{n,t})^{-\alpha-2}$$

θ - α

$$\frac{\partial^2 \ell}{\partial \theta \partial \alpha} = \sum_{n=1}^{N_T} \frac{l_n}{y_{n,l_n}} - A \sum_{n,t} t(z_{n,t})^{-\alpha-1} + \alpha A \sum_{n,t} t(z_{n,t})^{-\alpha-1} \log(z_{n,t})$$

A - α

$$\frac{\partial^2 \ell}{\partial A \partial \alpha} = -\frac{N_T}{A} + \sum_{n,t} y_{n,t} (z_{n,t})^{-\alpha-1} [1 - \alpha \log(z_{n,t})]$$